

Charles Zhao

Mechatronics Engineering Student · University of Waterloo

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EDUCATION

University of Waterloo

Sep 2023 – Expected Apr 2028

Honours Mechatronics Engineering, Co-op · Third year

TECHNICAL SKILLS

Electronics: Altium Designer, KiCad, analog front ends, power regulation, PCB assembly and testing

Embedded: C/C++, STC8/8051, ESP32, STM32, Keil C51, UART/SPI/I²C, RS485, Modbus

Software & tools: TypeScript, React, Electron, Node.js, Python, Git, SolidWorks, 3D printing

EXPERIENCE

Minebea Intec GmbH

Jun 2026 – Jul 2026

Embedded Hardware Engineering Intern

- Designed and developed the original PR5170 weighing instrument and its variants across two internships, spanning PCB hardware, embedded firmware, and desktop software.
- Designed or modified more than 20 PCBs across two work periods at Minebea Intec in 2025 and 2026.
- Designed PR5170 standard and Ex ec digital mainboards, plus the Ex ec analog mainboard (gas Zone 2).
- Adapted two Embedded C firmware branches for STC8/8051 acquisition, calibration, filtering, and RS485.
- Developed TConfig in Electron, React, and TypeScript for monitoring, parameter configuration, and calibration across multiple instruments, including PR5170.

Charles Tech

Jul 2025 – Mar 2026

Embedded Systems Engineer & Owner

- Delivered PCB design services for global clients, translating requirements into manufacturing-ready schematics and high-speed layouts.
- Established the engineering workflow and shifted the business toward proprietary electronic product development.

Minebea Intec GmbH

Jan 2025 – Apr 2025

PCB Designer

- Designed the PR5170 standard analog acquisition mainboard and shared power and industrial-interface baseboard, alongside HMI and protection-board variants.
- Integrated ADC and MCU circuitry, regulated supplies, RS232/RS485 interfaces, and 4–20 mA output functions.

Ningbo Chengzi Network Technology Co., Ltd

May 2024 – Aug 2024

Website Developer

- Developed a WeChat Mini Program prototype that attracted more than 100 early users in its first month.
- Improved LLM workflows through prompt engineering and context management; analyzed more than 10 competing products.

01 / ASSEMBLED HARDWARE

PR5170 hardware assembly example

Selected work from more than 20 PCBs designed or modified across two work periods at Minebea Intec in 2025 and 2026.



Shared baseboard and analog-input mainboard assembled with an industrial Ethernet module purchased by Minebea Intec.

Three-board assembly

Shared baseboard

Provides common power and industrial-interface connections.

Analog-input mainboard

Receives analog sensor signals for measurement and control.

Industrial Ethernet module

Purchased by Minebea Intec and integrated with the two boards.

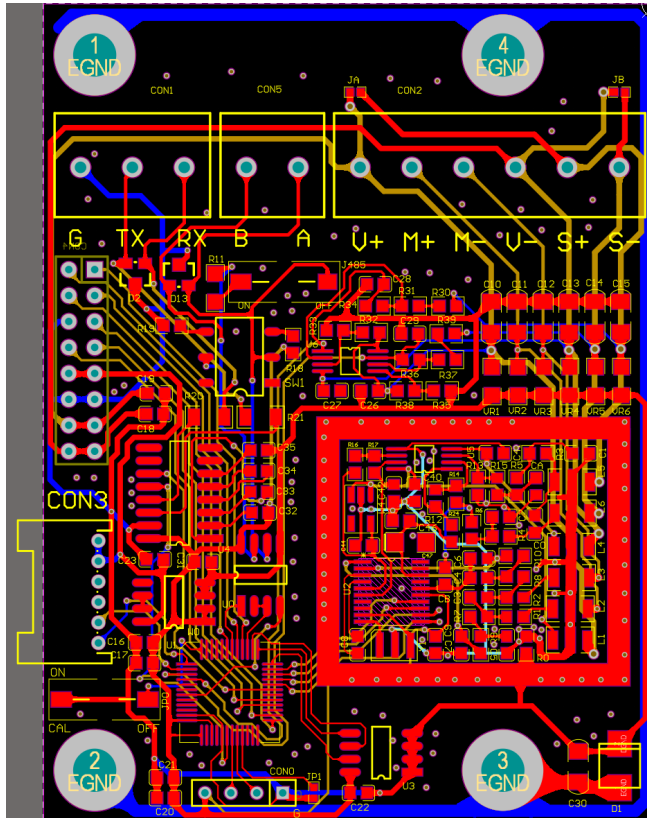
Proprietary details have been obfuscated for confidentiality. Selected works displayed with permission.

02 / PCB LAYOUT VARIANTS

Analog-input mainboard layouts

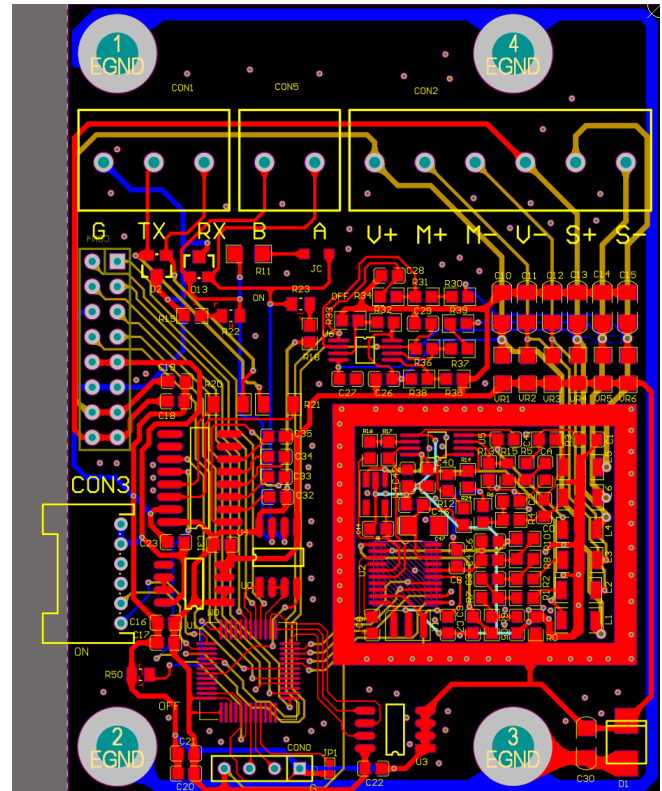
Standard-environment mainboard designed in 2025; Ex ec variant designed in 2026 for gas Zone 2 applications.

Standard-environment version



2025 / Standard environment. Load-cell acquisition with excitation and reference conditioning, controller logic, and industrial communication.

Gas Zone 2 version · Ex ec



2026 / Ex ec variant. Analog signal acquisition and control for gas Zone 2 instrument configurations.

Protection: Ex ec | **EPL:** Gc | **Application:** Gas Zone 2

Four- and six-wire sensor connections

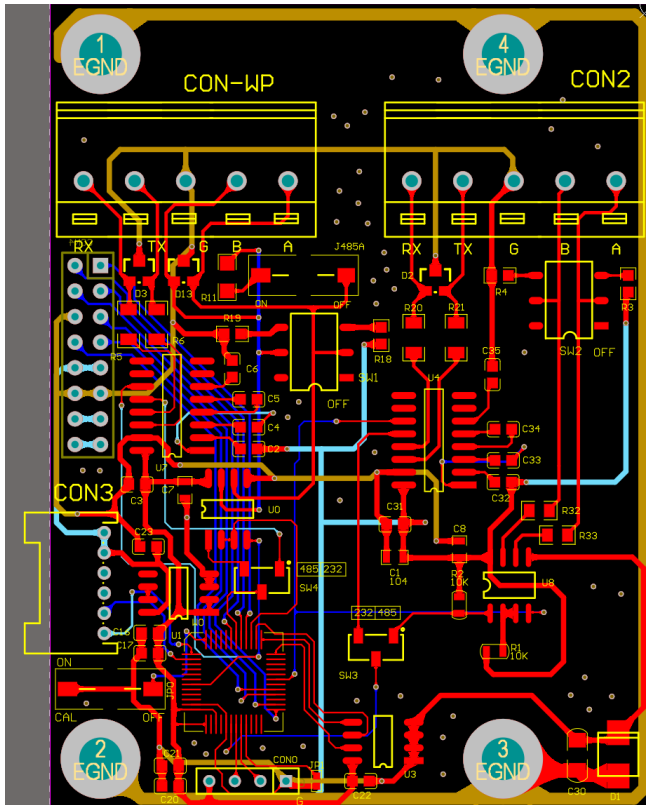
In the documented analog design, six-wire remote sensing lets a ratiometric reference follow excitation at the sensor. Four-wire connections use local sense jumpers. Symmetrical input filtering preserves the differential measurement path.

03 / PCB LAYOUT VARIANTS

Digital mainboard layouts

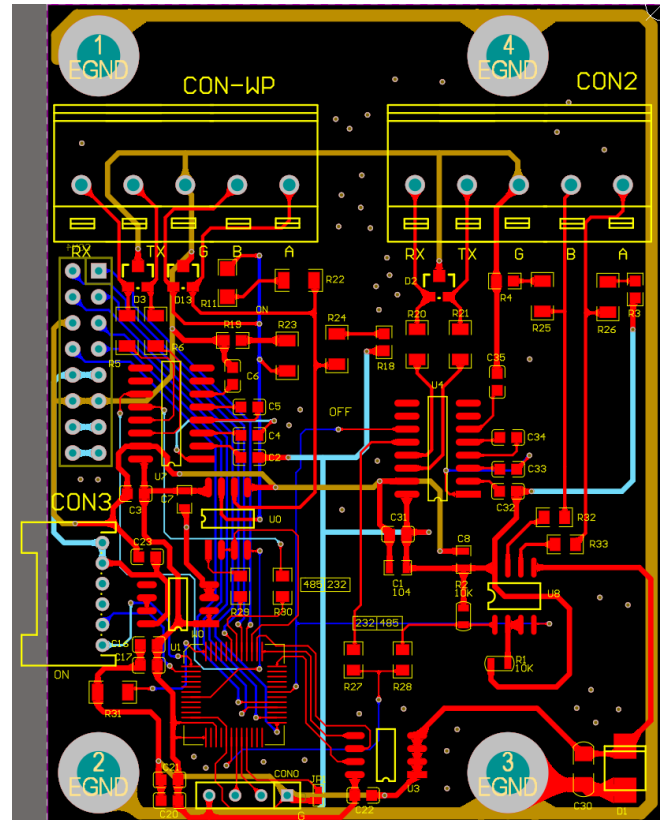
Designed in 2026 for external digital weighing platforms: a standard-environment version and an Ex ec version for gas Zone 2 applications.

Standard-environment version



2026 / Standard environment. Controller and serial interfaces receive already-digitized weighing data from an external platform.

Gas Zone 2 version · Ex ec



2026 / Ex ec variant. The same digital-platform role in a mainboard layout for gas Zone 2 instrument configurations.

Protection: Ex ec | **EPL:** Gc | **Application:** Gas Zone 2

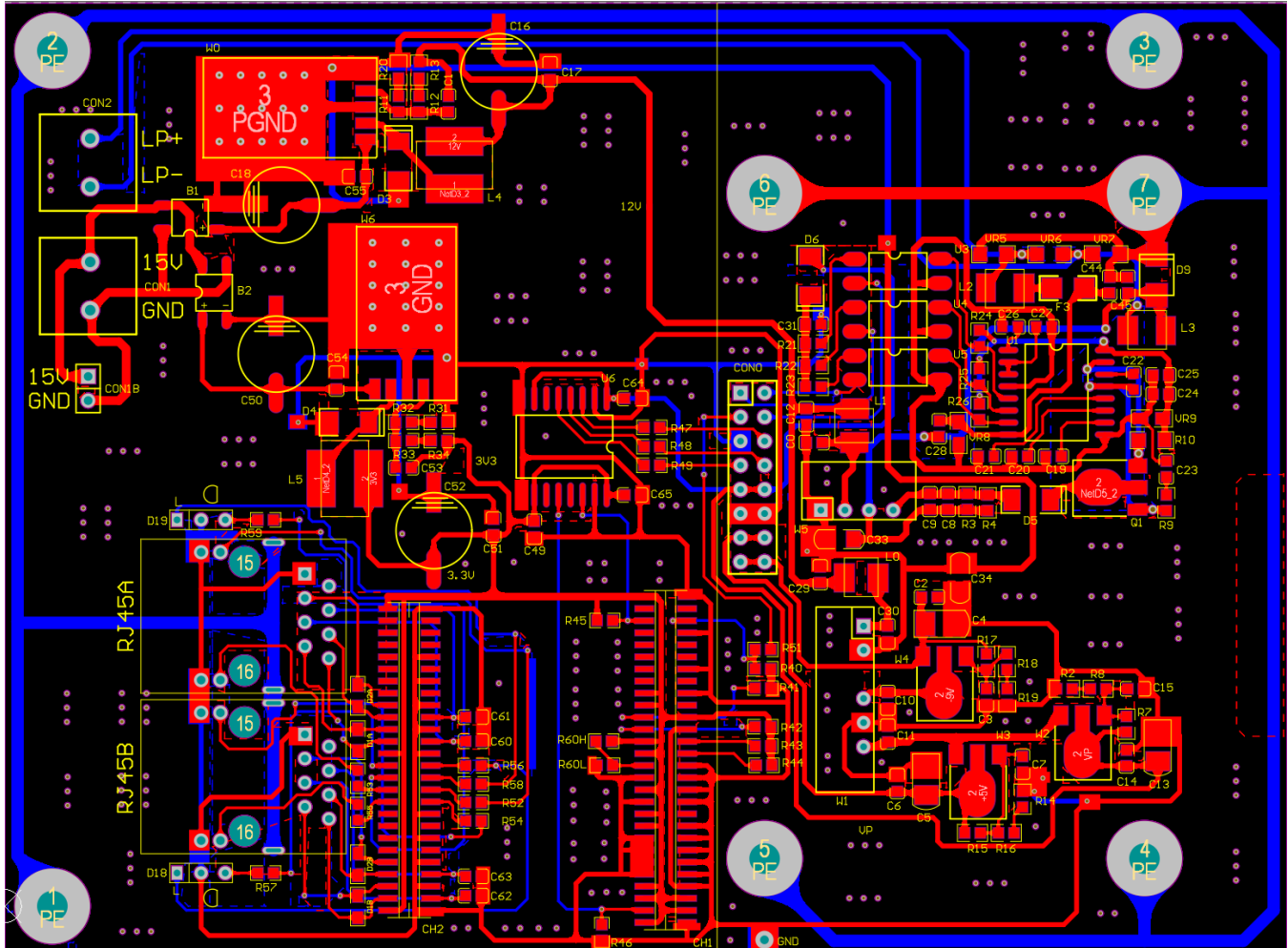
Two serial channels, two direction-control methods

Each channel exposes RS232 and RS485, with mechanical selection of the UART receive source. One RS485 channel uses MCU-controlled direction; the other uses hardware AutoDirection. Receive selection does not disconnect both transmit paths.

04 / POWER AND INDUSTRIAL INTERFACES

Shared baseboard / Type 1

Designed in 2025: a shared 24 V power and industrial-interface baseboard for both mainboard types.



PCB layout - Baseboard Type 1

Multi-rail power, connections for a company-purchased industrial Ethernet module, and an isolated 4-20 mA output.

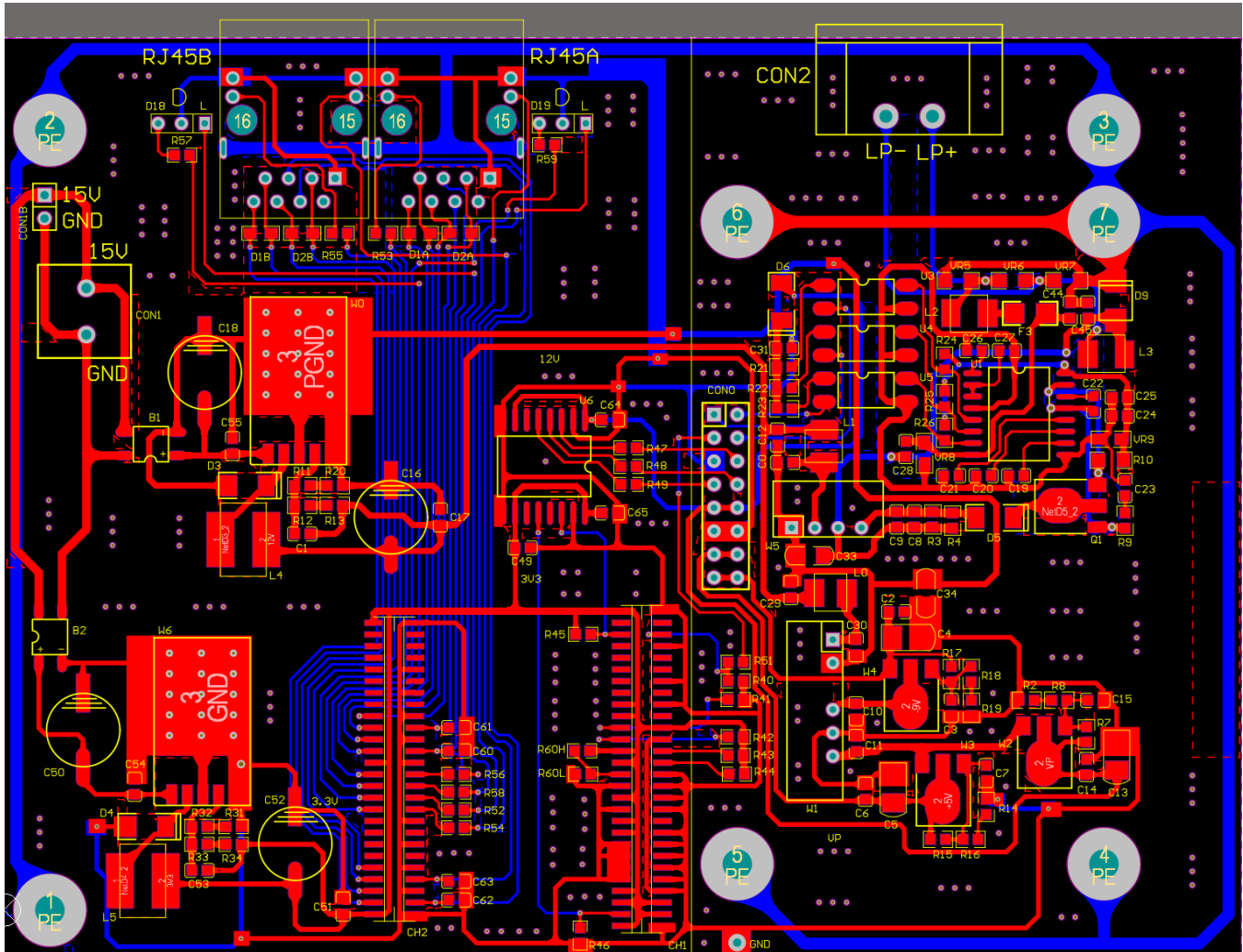
Keep the isolation boundary explicit

Optocouplers and an isolated DC/DC supply isolate the current output. In the documented circuit, the common-ground level-conversion path is separate from this isolation boundary.

04 / POWER AND INDUSTRIAL INTERFACES

Shared baseboard / Type 2

An alternative placement of the 2025 shared baseboard for a different mechanical enclosure.



PCB layout - Baseboard Type 2

Identical electrical functions to Type 1; component placement changes to fit a different enclosure.

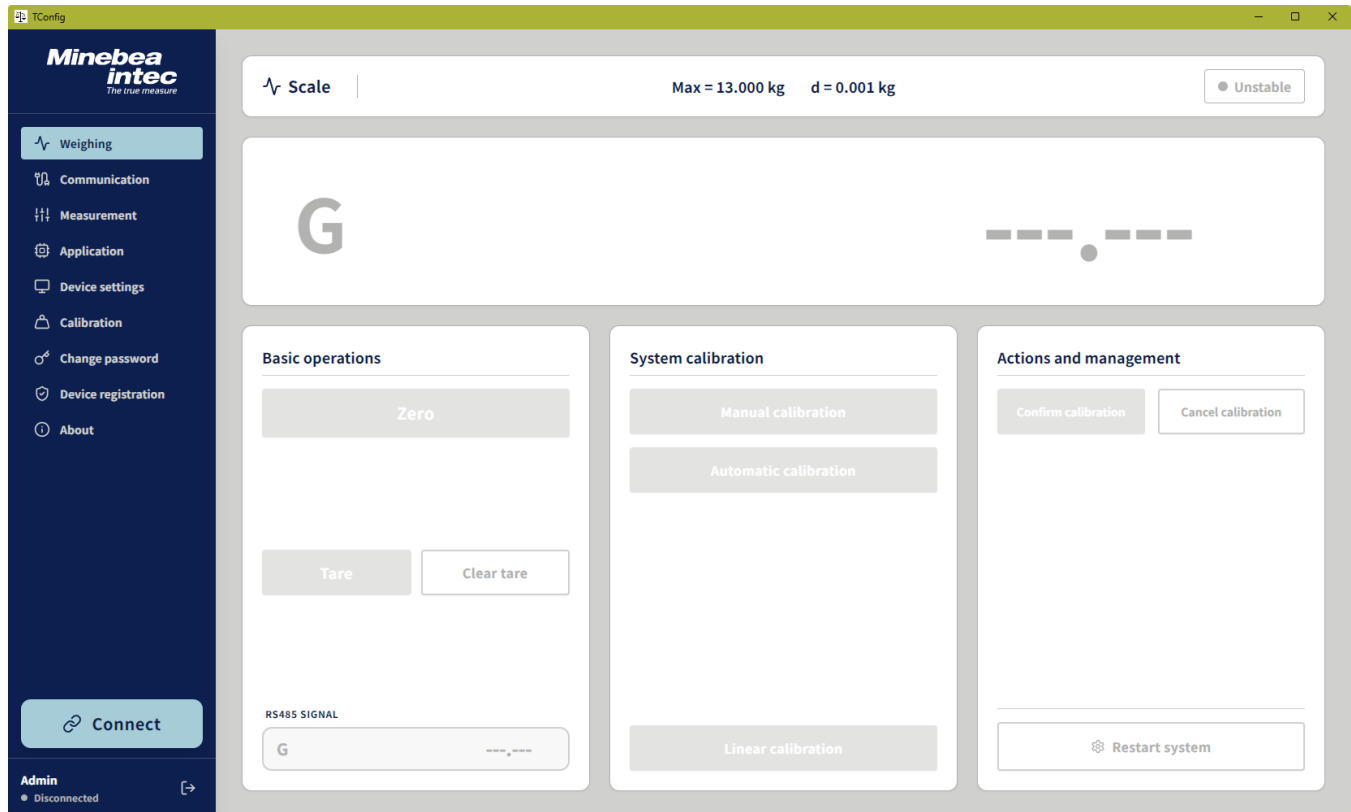
Reuse the electrical design

Both layouts support the same supply rails, industrial Ethernet module connections, and current-output function. This variant changes the mechanical arrangement while retaining compatibility with the mainboard interface.

05 / DESKTOP APPLICATION

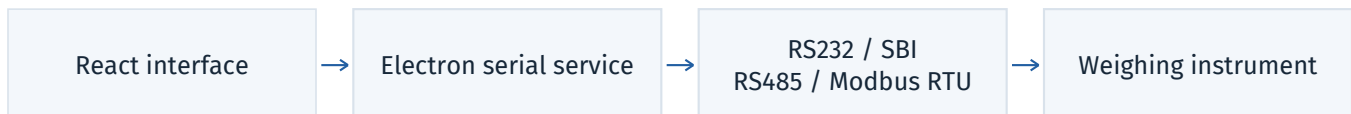
TConfig / Instrument workflows

I designed and developed TConfig with Electron, React, and TypeScript: a Windows application for monitoring, configuration, and calibration across multiple instruments, including PR5170.



Weighing workspace with weight, status, basic operations, and access to calibration controls.

Offline interface shown; placeholders are not live instrument readings.



Connect the interface to the instrument

A preload/IPC bridge connects the React workflows to the Electron serial service. The service assembles frames, matches replies to queued requests, checks Modbus CRCs, and handles timeouts.

Coordinate monitoring and parameter operations

Page state coordinates weight polling with parameter requests so delayed replies are handled in the correct workflow. Serial setup, measurement settings, device information, and calibration share one application.

06 / PARAMETER EDITING AND VERIFICATION

TConfig / Measurement settings

Advanced parameter workflows distinguish what the operator is editing from values confirmed by the instrument.

Basic measurement and advanced control parameters in one workspace, with separate read and set actions.

Offline interface shown; displayed values are not confirmed device readback.

Keep edited and confirmed values separate

Advanced forms protect fields being edited and retain a separate record of confirmed instrument values. A delayed reply should not silently replace the operator's in-progress changes.

Write in order, then read back

Selected writes run in sequence, followed by delayed readback and comparison. Field-level feedback reports mismatches or failures instead of treating transmission alone as confirmation.

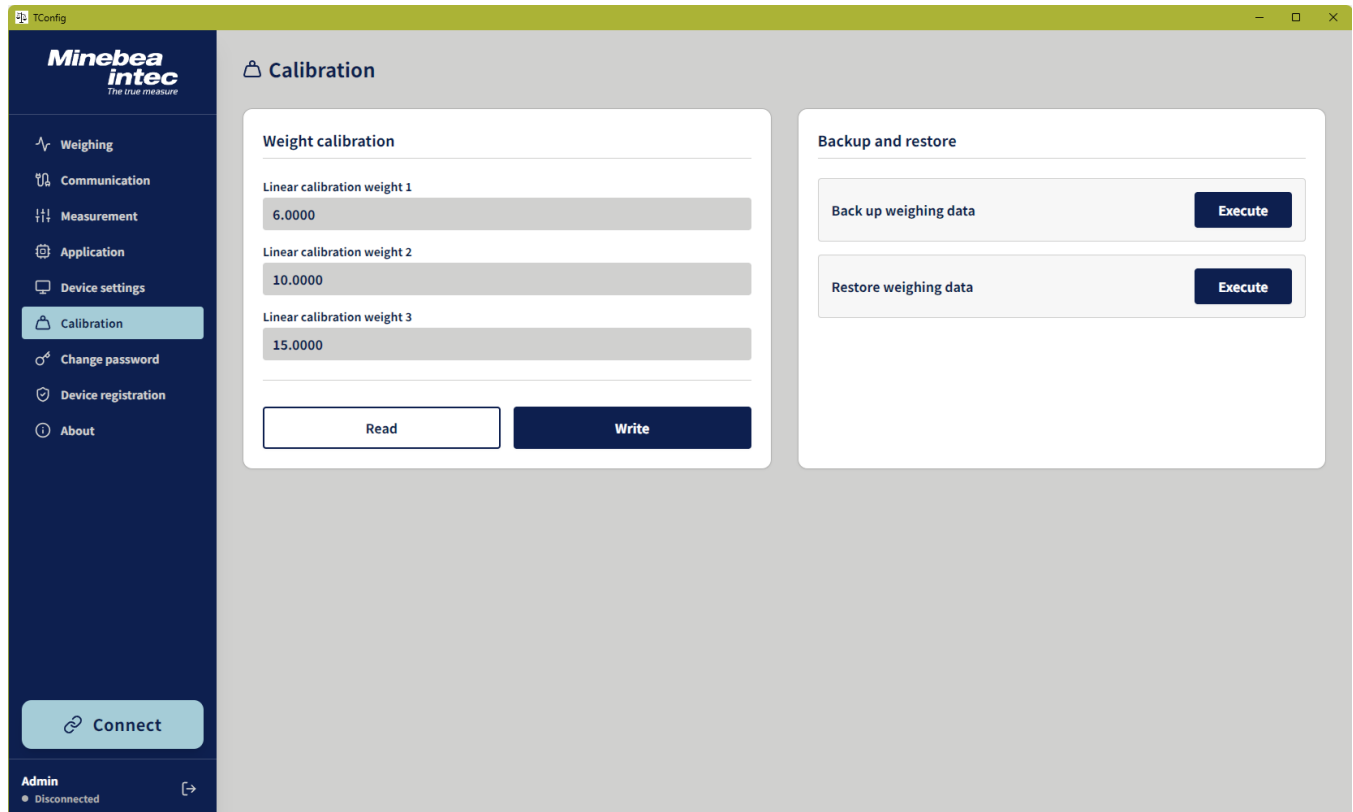
Make the evidence visible

Read and write feedback distinguishes interface state from device-confirmed state. Calibration-weight writes use their own numerical readback checks rather than assuming every parameter has the same completion behavior.

07 / READ BOUNDARIES AND WINDOWS DELIVERY

TConfig / Calibration controls

Read calibration-weight parameters without entering calibration mode, and keep device operations distinct from desktop interface actions.



Calibration-weight parameters and instrument backup/restore commands in the application interface.

Offline interface shown; displayed values are not evidence of completed calibration.

Passive reads preserve the instrument's operating mode

Passive queries, a read-only Modbus fallback, and shared requests retrieve calibration parameters without changing modes. Backup/restore sends device commands; instrument firmware performs the calibration.

Focused regression evidence

Seven simulated-serial tests cover read commands, fallback, unavailable channels, command restrictions, and concurrent requests.

Windows delivery

Delivered nine application pages plus sign-in and a Windows x64 installer with native serial support. Protocol documents, serial traces, and instrument feedback guided development; TypeScript checks passed.